

NammaPG – Room Entity: Complete Design & PG Relationship (PDF-5 Revised)

This document provides the complete and final design of the Room entity in the NammaPG backend system. Rooms represent the physical living units inside a PG and are responsible for sharing type, capacity, and availability management.

Why Room is a Separate Entity

A PG contains multiple rooms with different sharing configurations and capacities. Room-level details must not be stored in the PG table. Separating Room ensures clean normalization, flexibility, and scalability.

Room Entity – Final Schema

id – Primary key
roomNumber – Unique room identifier within a PG
floorNumber – Floor on which the room is located
sharingType – SINGLE / DOUBLE / TRIPLE (stored as String)
capacity – Total number of beds in the room
availableBeds – Current available beds
active – Soft enable/disable flag
pg_id – Foreign key reference to PG
createdAt – Record creation timestamp
updatedAt – Record update timestamp

Sharing & Bed Logic

Room sharing type and bed count are handled directly within the Room table. There is no separate RoomType or Bed table in Phase-1. Capacity represents total beds, while availableBeds tracks live availability. This design avoids overengineering while remaining scalable.

PG–Room Relationship

One PG can have many rooms, and each room belongs to exactly one PG. This is implemented using a One-to-Many relationship from PG to Room and a Many-to-One relationship from Room to PG. Room is the owning side.

What Room Does NOT Contain

Room does not store pricing, facilities, booking details, or user information. These concerns are handled by separate tables such as Pricing, Facility, and Booking. This separation ensures clean responsibility boundaries.

Design Stability

This Room design is stable, production-ready, and interview-safe. Future enhancements such as pricing models or per-bed booking can be added without breaking the existing schema.

Next Tables

The next entities to be designed are Facility (with PG-Facility mapping), Booking (User-Room relationship), and Issue/Complaint tracking.